

On Inertia Distribution, Inter-Area Oscillations and Location of Electronically-Interfaced Resources

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Abstract—This paper explores the relationships between inertia distribution, inter-area oscillations, and location of electronically-interfaced resources that are enabled with either damping or inertia emulation controllers (EIRs). A two-machine system with an EIR is used for analytical derivations. Explicit analytical expressions are found for: (a) the location of the center of inertia (COI), which depends on the H-inertia constant and voltage set-points of the machines, and (b) the residue of the system transfer function, which is convex in terms of the EIR location. These expressions are validated using full-order models for machines, exciters, and governors; the results support the idea of placing EIRs further away from the COI, or equivalently, in areas with low inertia to attain the highest possible oscillation damping. In the case of large-scale systems, an inertia distribution index is proposed which allows estimating the distance from any bus to the COI location. The efficacy of the proposed index is tested in a real system. The system areas with less inertia are proved to be the best places to deploy EIRs. The proposed index does not require computationally expensive calculations as those from modal analysis and it seems promising to better reinforce power system dynamics through EIRs.

Index Terms—Energy storage, power system stability, power system control, power system planning, wind power generation.

I. INTRODUCTION

POWER systems are moving towards a more renewable generation mix motivated by critical issues such as environmental pollution and fossil-fuel dependency. Although being imperative, this system transformation is creating technical concerns and, furthermore, as renewable generating units behave so differently from conventional synchronous machines, fundamental changes in the way power systems are operated and planned are required. One of these concerns is power system robustness. A relative reduction of system inertia, and the inherent variability of renewable energy generation may create larger frequency excursions and set the conditions for a more recurrent appearance of low frequency oscillations. To overcome this problem, additional regulation may be employed from

electronically-interfaced resources such as non-conventional renewable generating systems (NRGS) or energy storage systems (ESS).

For NRGS such as wind farms or ESS such as flywheels, control actions to provide additional damping or emulated inertia can be enabled to support power systems during frequency excursions and oscillations [1]–[4]. Control design in both NRGS and ESS is very similar and based on either phase compensators, filters or derivative blocks [5]–[7]; the main difference has to do with the time-variant behavior of these control actions in wind farms due to wind variability [8]. These control actions have also been studied using HVDC links [9]–[11]. An early study on this subject compares active and reactive power modulation of an HVDC link in a so-called perpendicular system [11]—a 2-machine equivalent system tapped in between by an HVDC link. The power modulation is done using frequency measurements from both synchronous machines. By using eigenvalue sensitivity analysis, and given a specific point of connection of the HVDC link, the authors conclude that active power modulation is more effective when only the frequency of the machine at the shortest inertia-scaled electrical distance is employed. On the other hand, reactive power modulation is found to be more effective at high power transfers and the frequency of the power receiving machine should be employed for modulation. These conclusions are drawn based on the direction of the eigenvalue sensitivity with respect to the modulation gains. Although these are interesting results, more emphasis should have been placed on the magnitude of the sensitivity, as the direction of eigenvalue displacement can be corrected by using phase compensation blocks. We recognize that by locating properly not just HVDC links but any other EIR, the magnitude of the eigenvalue sensitivity with respect to a controller gain can be increased, and by using a proper phase compensation, power system dynamics can be considerably improved [4], [12]. In this paper, note that the term EIRs is used exclusively for electronically-interfaced resources enabled with either damping or inertia emulation controllers.

Optimal placement of EIRs is relatively a new area of research. Some work has been done by formulating the problem as the maximization of synchronous machine coherency after perturbations; unfortunately, the conclusion obtained so far is somehow trivial: the best place corresponds to the point of perturbation [13]. Another attempt to understand the location problem has also been reported considering a wind farm in a radial system [14]. The problem was formulated using a spatial-temporal model based on a continuous sequence of classically represented synchronous generators. While a few locations are

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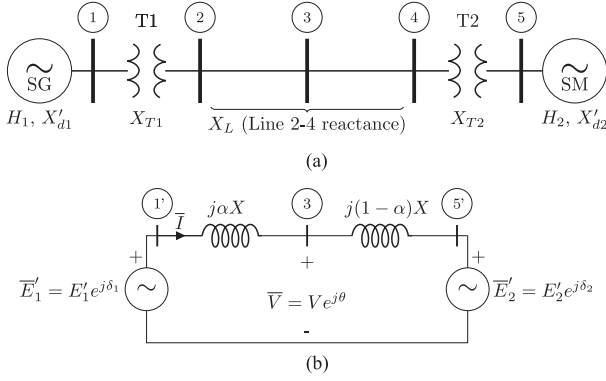


Fig. 1. 2-machine system: (a) one-line diagram, (b) equivalent circuit.

evaluated along this radial system, the wind farm has no close-loop controller and its impact on the system dynamics is merely related to changes in the system operating point (voltages and power flows); no conclusive results in regards to the ideal wind farm location are reported.

This paper provides new grounds on the subject of locating EIRs. By using a 2-machine test system, the fundamental principles to place EIRs are laid down, and upon them, an index of inertia distribution is proposed for locating optimally these resources in large scale power systems. In the 2-machine system, analytical derivations are done to obtain: (a) an explicit mathematical expression for the COI location, which depends on machine H-inertia constants and voltage controller set-points, and (b) an explicit mathematical expression for the system transfer function residue, which is convex in terms of the location and exhibits a minimum value exactly at the COI. By these derivations, concrete proofs are provided to conclude that these resources should be located at the furthest distances from the COI. In the case of large scale systems, as analytical expressions are prohibited and undetermined, an index is proposed that is able to capture the distance of a particular bus to the COI—related as well to the inertia distribution in the system. By using a real power system, this index is validated by comparing its values against the residue calculated using modal analysis of the system full order model. A highly linear correlation is found between the index and the residue; thus, rather than performing computationally expensive calculations through modal analysis, a simple notion of the distribution of inertia in the system can be used for planning EIRs deployment. This paper is structured as follows. In Section II, analytical derivations for both the COI location and the transfer function residue in a 2-machine test system are presented. In Section III, based on the derivations using the 2-machine system, an index to locate EIRs in large scale systems is proposed and tested in a real system model. Conclusions are presented in Section IV.

II. DISTRIBUTION OF INERTIA AND OSCILLATIONS: 2-MACHINE SYSTEM CASE

Consider the lossless system shown in Fig. 1. A synchronous generator (SG) and a synchronous motor (SM) are connected to buses 1 and 5, respectively. Assume that both machines are represented by their classical model, i.e., internal transient voltage $\bar{E}'_i$ in series with the d-axis transient reactance X'_{di} . A generic

bus 3 is defined at some point within the line 2–4 to explore how voltage angle (θ) and frequency (ω) change along the line. Define the location parameter α as the relative distance from bus 3 to the generator internal transient voltage (node 1'). Thus,

$$0 \leq \alpha_{\min} \leq \alpha \leq \alpha_{\max} \leq 1 \quad (1)$$

where $\alpha_{\min} = (X'_{d1} + X_{T1})/X$, $\alpha_{\max} = (X - X'_{d2} - X_{T2})/X$ and $X = X'_{d1} + X_{T1} + X_L + X_{T2} + X'_{d2}$ is the circuit total reactance. The system transient response is mathematically represented by the following initial value problem:

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\delta}_1 \\ \dot{\delta}_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2H_1} \left(P_{m1} - \frac{E'_1 E'_2}{X} \sin(\delta_1 - \delta_2) \right) \\ \frac{1}{2H_2} \left(\frac{E'_1 E'_2}{X} \sin(\delta_1 - \delta_2) - P_{m2} \right) \\ \omega_s (\omega_1 - 1) \\ \omega_s (\omega_2 - 1) \end{bmatrix} \quad (2)$$

$$\begin{bmatrix} \omega_1(t_0) \\ \omega_2(t_0) \\ \delta_1(t_0) \\ \delta_2(t_0) \end{bmatrix} = \begin{bmatrix} \omega_{10} \\ \omega_{20} \\ \delta_{10} \\ \delta_{20} \end{bmatrix} \quad (3)$$

All variables and parameters are in per unit, except for ω_s measured in radians per second, H_1 and H_2 in seconds, δ_1 and δ_2 in radians. Note that both transient internal voltages and the mechanical power are calculated based on the pre-disturbance condition (equilibrium point) and they are assumed to remain fixed during the transient response, i.e., no automatic voltage regulator and governor are considered. The disturbance is applied through the initial value defined in (3). For now, bus 3 has no impact on the system dynamic and it is only considered as a point to observe angle and frequency along the line. An explicit analytical expression for θ in terms of the state variables is found as follows. Apply voltage Kirchhoff's law to obtain:

$$V e^{j\theta} = E'_1 e^{j\delta_1} - j\alpha X \underbrace{\left(\frac{E'_1 e^{j\delta_1} - E'_2 e^{j\delta_2}}{jX} \right)}_{\bar{I}} \quad (4)$$

$$\Rightarrow V = E'_1 e^{j(\delta_1 - \theta)} - \alpha (E'_1 e^{j\delta_1} - E'_2 e^{j\delta_2}) e^{-j\theta} \quad (5)$$

$$= (1 - \alpha) E'_1 e^{j(\delta_1 - \theta)} + \alpha E'_2 e^{j(\delta_2 - \theta)} \quad (6)$$

Thus, by taking the imaginary part:

$$0 = (1 - \alpha) E'_1 \sin(\delta_1 - \theta) + \alpha E'_2 \sin(\delta_2 - \theta) \quad (7)$$

$$\approx (1 - \alpha) E'_1 (\delta_1 - \theta) + \alpha E'_2 (\delta_2 - \theta) \quad (8)$$

Here, the angle differences are assumed to be small enough such that $\sin(\delta_1 - \theta) \approx \delta_1 - \theta$ and $\sin(\delta_2 - \theta) \approx \delta_2 - \theta$. By solving for θ in (8), the following explicit relationship is obtained:

$$\theta = \frac{(1 - \alpha) E'_1}{(1 - \alpha) E'_1 + \alpha E'_2} \delta_1 + \frac{\alpha E'_2}{(1 - \alpha) E'_1 + \alpha E'_2} \delta_2 \quad (9)$$

and the angle time derivative becomes:

$$\dot{\theta}(\alpha) = \frac{(1-\alpha)E'_1\omega_s(\omega_1-1)}{(1-\alpha)E'_1 + \alpha E'_2} + \frac{\alpha E'_2\omega_s(\omega_2-1)}{(1-\alpha)E'_1 + \alpha E'_2} \quad (10)$$

$$= \omega_s(\omega-1) \quad (11)$$

where

$$\omega = \frac{(1-\alpha)E'_1}{(1-\alpha)E'_1 + \alpha E'_2}\omega_1 + \frac{\alpha E'_2}{(1-\alpha)E'_1 + \alpha E'_2}\omega_2 \quad (12)$$

corresponds to the frequency in per unit at bus 3.

A. Center of Inertia Formula

By adjusting the location parameter, there exists an α^* such that bus 3 is located exactly at the system COI. A physical condition for the COI is that its changes in angle (θ) and frequency (ω) are minimal when compared to the angle/frequency of any other bus in the system. Thus, the COI location is formally given by:

$$\alpha^* = \arg \min_{\alpha \in [\alpha_{\min}, \alpha_{\max}]} \dot{\theta}(\alpha)^2 \quad (13)$$

By Karush-Kuhn-Tucker optimality condition:

$$\frac{d\{\dot{\theta}(\alpha^*)^2\}}{d\alpha} = 0 = 2\dot{\theta}(\alpha^*)\omega_s \frac{E'_1 E'_2 (\omega_2 - \omega_1)}{\{(1-\alpha^*)E'_1 + \alpha^* E'_2\}^2} \quad (14)$$

$$\Rightarrow \dot{\theta}(\alpha^*) = 0 \quad (15)$$

Note that (14) also vanishes when $\omega_1 = \omega_2$, but this does not hold for all time in $[t_0, \infty)$. Assuming stability, the condition from (15) can be alternatively enforced using $\dot{\theta}(\alpha^*) = 0 \forall t \in [t_0, \infty)$. Thus,

$$\ddot{\theta}(\alpha^*) = 0 = \frac{(1-\alpha^*)E'_1\omega_s}{(1-\alpha^*)E'_1 + \alpha^* E'_2} \frac{d\omega_1}{dt} + \frac{\alpha^* E'_2\omega_s}{(1-\alpha^*)E'_1 + \alpha^* E'_2} \frac{d\omega_2}{dt} \quad (16)$$

By using the equation of motion of both machines and as $P_{m1} = P_{m2}$, the optimality condition becomes:

$$\ddot{\theta}(\alpha^*) = 0 = \frac{\omega_s}{2} \left[\underbrace{P_{m1} - \frac{E'_1 E'_2}{X} \sin(\delta_1 - \delta_2)}_{=0 \text{ in steady-state only}} \right] \left[\frac{(1-\alpha^*)\frac{E'_1}{H_1} - \alpha^*\frac{E'_2}{H_2}}{(1-\alpha^*)E'_1 + \alpha^* E'_2} \right] \quad (17)$$

$$\Rightarrow 0 = (1-\alpha^*)\frac{E'_1}{H_1} - \alpha^*\frac{E'_2}{H_2} \quad (18)$$

$$\Rightarrow \alpha^* = \frac{E'_1 H_2}{E'_1 H_2 + E'_2 H_1} = \frac{1}{1 + \frac{E'_2}{E'_1} \frac{H_1}{H_2}} \quad (19)$$

A physical COI location is found, α^* , which depends on both inertia and voltage of the machines. Observe that E'_1 and E'_2 are around the machine voltage set-points; although adjustable during operation, they are very similar to each other and are close to one per unit. On the other hand, H_1 and H_2 are not subject to operational adjustments but are fixed quantities related to machine physical characteristics; besides, depending on the

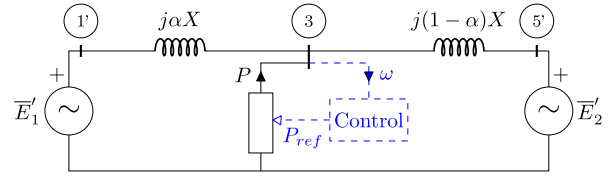


Fig. 2. 2-machine system with an EIR at bus 3.

technology and size, their values may differ considerably from each other. In summary, the COI location is found to depend strongly on the inertia ratio H_1/H_2 and slightly on the voltage ratio E'_2/E'_1 . It is worth to emphasize that, in the literature, the well-known COI frequency has been used for modeling and analysis of power systems, but its location remained undetermined [15], [16]. For completeness, by replacing (19) into (12), the COI frequency becomes:

$$\omega^* = \frac{H_1\omega_1 + H_2\omega_2}{H_1 + H_2} \quad (20)$$

B. System Dynamics Including an EIR

An EIR is connected to the generic bus 3 (see Fig. 2). For the sake of simplicity, the EIR reactive power is assumed to be zero and its active power is modulated within a small range compared to the power exchange between machines. With the inclusion of the EIR, the new relationships among voltage at bus 3 (V , θ) and the state variables are found as follows:

$$S_3 = V e^{j\theta} \text{conj} \left\{ \underbrace{\frac{V e^{j\theta} - E'_1 e^{j\delta_1}}{j\alpha X} + \frac{V e^{j\theta} - E'_2 e^{j\delta_2}}{j(1-\alpha)X}}_{\bar{I}_3} \right\} \quad (21)$$

$$= \frac{jV^2}{\alpha(1-\alpha)X} - \frac{jV E'_1 e^{j(\theta-\delta_1)}}{\alpha X} - \frac{jV E'_2 e^{j(\theta-\delta_2)}}{(1-\alpha)X} \quad (22)$$

where $\bar{I}_3$ is the complex current injected to bus 3 and $S_3 = P$ is purely real. By taking real and imaginary parts of (22), two algebraic equations are obtained. As a result, the open-loop system dynamics are represented by the following set of differential-algebraic equations (DAEs):

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \\ \dot{\delta}_1 \\ \dot{\delta}_2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2H_1} \left(P_{m1} - \frac{E'_1 V}{\alpha X} (\delta_1 - \theta) \right) \\ \frac{1}{2H_2} \left(\frac{V E'_2}{(1-\alpha)X} (\theta - \delta_2) - P_{m2} \right) \\ \omega_s (\omega_1 - 1) \\ \omega_s (\omega_2 - 1) \\ -P + \frac{V E'_1 (\theta - \delta_1)}{\alpha X} + \frac{V E'_2 (\theta - \delta_2)}{(1-\alpha)X} \\ \frac{V^2}{\alpha(1-\alpha)X} - \frac{V E'_1}{\alpha X} - \frac{V E'_2}{(1-\alpha)X} \end{bmatrix} \quad (23)$$

Once again, angles differences are assumed to be small enough. The system output is the frequency at bus 3, which is used for controlling P . For simplification, and as P is smaller than the power exchanged by the machines, the frequency at bus 3 can be fairly approximated by (12). At the equilibrium point, given P_{m2} , SG and EIR power must be chosen to satisfy $P_{m1} + P = P_{m2}$.

Consider the vectors $x = [\omega_1 \ \omega_2 \ \delta_1 \ \delta_2]^T$, $y = [V \ \theta]^T$ and $P_m = [P_{m1} \ P_{m2}]^T$. As P varies for regulation purposes in a small range around the equilibrium point, the model is linearized to obtain:

$$\begin{bmatrix} \Delta \dot{x} \\ 0 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} + \begin{bmatrix} B_1 \\ B_3 \end{bmatrix} \Delta P_m + \begin{bmatrix} B_2 \\ B_4 \end{bmatrix} \Delta P \quad (24)$$

$$\Delta \omega = \begin{bmatrix} C_1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \quad (25)$$

Algebraic equations are eliminated using Kron's reduction as:

$$\begin{aligned} \Delta \dot{x} &= (A_1 - A_2 A_4^{-1} A_3) \Delta x + (B_1 - A_2 A_4^{-1} B_3) \Delta P_m \\ &\quad + (B_2 - A_2 A_4^{-1} B_4) \Delta P \end{aligned} \quad (26)$$

As B_2 and B_3 are null matrices, the linearized model becomes:

$$\Delta \dot{x} = A_\alpha \Delta x + B_1 \Delta P_m + B_\alpha \Delta P \quad (27)$$

$$\Delta \omega = C_\alpha \Delta x \quad (28)$$

where,

$$A_\alpha = A_1 - A_2 A_4^{-1} A_3 = \begin{bmatrix} 0 & 0 & -\frac{n}{H_1} & \frac{n}{H_1} \\ 0 & 0 & \frac{n}{H_2} & -\frac{n}{H_2} \\ \omega_s & 0 & 0 & 0 \\ 0 & \omega_s & 0 & 0 \end{bmatrix} \quad (29)$$

$$B_\alpha = -A_2 A_4^{-1} B_4 = \begin{bmatrix} \frac{(1-\alpha)E'_1}{2H_1 m} & \frac{\alpha E'_2}{2H_2 m} & 0 & 0 \end{bmatrix}^T \quad (30)$$

$$C_\alpha = C_1 = \begin{bmatrix} \frac{(1-\alpha)E'_1}{m} & \frac{\alpha E'_2}{m} & 0 & 0 \end{bmatrix} \quad (31)$$

$$m = (1 - \alpha)E'_1 + \alpha E'_2 \quad (32)$$

$$n = \frac{V_0 E'_1 E'_2}{2Xm} \quad (33)$$

Here, V_0 is the bus 3 voltage magnitude at the equilibrium. A_α , B_α and C_α are the system matrix, the input coefficient vector and the output coefficient vector, respectively; the use of the subscript α is to emphasize on the location parameter dependency. To obtain an analytical expression for the electromechanical eigenvalue, this fourth order system must be first reduced to a second order representation. As shown in the Appendix, the electromechanical eigenvalue is given by:

$$\lambda_\alpha = j \sqrt{\frac{\omega_s V_0}{2X} \left(\frac{H_1 + H_2}{H_1 H_2} \right) \frac{E'_1 E'_2}{(1 - \alpha)E'_1 + \alpha E'_2}} \quad (34)$$

Let $v_\alpha = [a \ b \ c \ d]^T$ be the right eigenvector of λ_α , thus:

$$0 = [A - \lambda_\alpha I] v_\alpha \quad (35)$$

$$= \begin{bmatrix} -\lambda_\alpha & 0 & -\frac{n}{H_1} & \frac{n}{H_1} \\ 0 & -\lambda_\alpha & \frac{n}{H_2} & -\frac{n}{H_2} \\ \omega_s & 0 & -\lambda_\alpha & 0 \\ 0 & \omega_s & 0 & \lambda_\alpha \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \quad (36)$$

$$\Rightarrow H_1 a = -H_2 b \quad c = \frac{\omega_s}{\lambda_\alpha} a \quad d = \frac{\omega_s}{\lambda_\alpha} b \quad (37)$$

In a similar fashion, with $w_\alpha = [e \ f \ g \ h]^T$ being the left eigenvector of λ_α , it can be proved that:

$$e = -f \quad g = \frac{\lambda_\alpha}{\omega_s} e \quad h = \frac{\lambda_\alpha}{\omega_s} f \quad (38)$$

Finally, by setting $a = H_2$ and $e = 1$, the right and left eigenvectors become:

$$v_\alpha = \begin{bmatrix} H_2 \\ -H_1 \\ \frac{\omega_s}{\lambda_\alpha} H_2 \\ -\frac{\omega_s}{\lambda_\alpha} H_1 \end{bmatrix} \quad w_\alpha = \begin{bmatrix} 1 \\ -1 \\ \frac{\lambda_\alpha}{\omega_s} \\ -\frac{\lambda_\alpha}{\omega_s} \end{bmatrix} \quad (39)$$

For verification purposes, note that the mode shape of the speed of machines 1 and 2, by inspection of (39), corresponds exactly to 0° and 180° , respectively.

C. Eigenvalue Sensitivity and Residue

The effectiveness of the EIR to damp the electromechanical oscillation is explored for all locations $\alpha \in [\alpha_{\min}, \alpha_{\max}]$ by using eigenvalue sensitivity. If the EIR is represented in the frequency domain by $K F(s)$, with K being the EIR gain and $F(s)$ a normalized transfer function, the sensitivity of the electromechanical eigenvalue (λ_α) with respect to the EIR gain is given by [17]:

$$\left| \frac{\partial \lambda_\alpha}{\partial K} \right| = \left| w_\alpha^T \frac{\partial A_\alpha}{\partial K} v_\alpha \right| = |R_\alpha| |F(\lambda_\alpha)| \quad (40)$$

where $R_\alpha = C_\alpha v_\alpha w_\alpha^T B_\alpha$ is known as the system transfer function residue associated with the mode e^{λ_α} . Note that for a given EIR with a specific controller to add either damping or inertia into the system, its transfer function does not change when a different location α is selected, except for the phase compensation blocks required to displace the eigenvalue towards the left half plane—note that a phase compensation would be needed when the residue R_α has an angle different than 180° . Therefore, independent of the type of controller used in the EIR, $|F(\lambda_\alpha)|$ remains unchangeable with respect to α , and the residue becomes the only index that provides a useful metric for selecting the EIR location: the larger the residue, the greater the impact on the electromechanical mode. By replacing (30), (31), (34)

and (39), the residue expression in terms of α becomes:

$$R_\alpha = \frac{H_1 H_2}{2} \left(\frac{(1-\alpha) \frac{E'_1}{H_1} - \alpha \frac{E'_2}{H_2}}{(1-\alpha) E'_1 + \alpha E'_2} \right)^2 \quad (41)$$

Thus, the residue is a nonnegative convex expression in terms of α . By simple inspection of (17) and (41), it is concluded that the minimum value of the residue is zero and occurs exactly at the COI location α^* . After some algebraic manipulations, the residue can be conveniently written as:

$$R_\alpha = \frac{H_1 H_2}{2} \frac{\left(\frac{E'_1}{H_1} + \frac{E'_2}{H_2} \right)^2 (\alpha - \alpha^*)^2}{(E'_1 + \alpha(E'_2 - E'_1))^2} \quad (42)$$

By assuming further that $|E'_1| \gg \alpha|E'_2 - E'_1|$, the residue expression is simplified to:

$$R_\alpha \approx \frac{H_1 H_2}{2 E_1'^2} \left(\frac{E'_1}{H_1} + \frac{E'_2}{H_2} \right)^2 (\alpha - \alpha^*)^2 \quad (43)$$

Note that the higher the distance $\alpha - \alpha^*$, the larger the residue R_α . In other words, in regards to inter-area oscillations, this explicit expression for the residue indicates that the best locations to deploy an EIR are those that are electrically farther away from the system COI. From this result, it turns out that a simple notion of the COI location of a system and a proper metric for the distance $\alpha - \alpha^*$ would suffice as a criterion to identify ideal locations for EIRs deployment. By this approach, computationally expensive calculations to obtain $R_\alpha = C_\alpha v_\alpha w_\alpha^T B_\alpha$, which are sensitive to operational conditions, can be avoided.

D. Validation Using Higher Order Models

The analytical expressions presented in (19) and (43) are tested against both the numerical values for the COI location and the residue obtained through full order time-domain simulations and modal analysis. To obtain the COI location numerically, the angle derivative square at all given locations, as stated in (13), is integrated over time after a 30 mHz initial deviation for ω_1 and ω_2 . The location with the minimal integral corresponds to the COI location. For the machines, the classical model (2nd order) and a 6th order model are considered for comparison purposes, as well as diverse exciters and governors. The system data (Fig. 1) is as follows: $P_{m1} = 20$ MW, $V_1 = 1.05$ p.u., $V_5 = 0.95$ p.u., $H_1 = H_2 = 4$ s, $X'_{d1} = X'_{d2} = 0.15$ p.u., $X_{T1} = X_{T2} = 0.06$ p.u., $X_L = 3.85$ p.u., all in base of 100 MVA. With these parameters, the minimum and maximum values for α are $\alpha_{\min} = 0.0492$ and $\alpha_{\max} = 0.9508$. The equilibrium point corresponds to $\omega_1 = \omega_2 = 1$ p.u. and $\delta_1 = 52.7^\circ$ and $\delta_2 = 0^\circ$. Note this is a high loading scenario and the linear assumptions made in Sections II-A and II-B are hard to justify; still, as shown next, the derived expressions for the COI location and residue are in good agreement with the exact values. DIGSILENT Power Factory and its model library have been employed for simulations. The obtained numerical results are shown in Table I. Note that the dynamics included by the machine 6th order model and the different exciters have minimal impact and the numerical results are very similar to the

TABLE I
NUMERICAL CALCULATION OF THE COI LOCATION
FOR THE 2-MACHINE TEST SYSTEM

Case	AVR	Machine	$\alpha^*_{(\text{numerical})}$
1	None	2nd	0.5275
2	None	6th	0.5298
3	SCRX	2th	0.5301
4	SCRX	6th	0.5306
5	IEEE T1	2nd	0.5297
6	IEEE T1	6th	0.5303

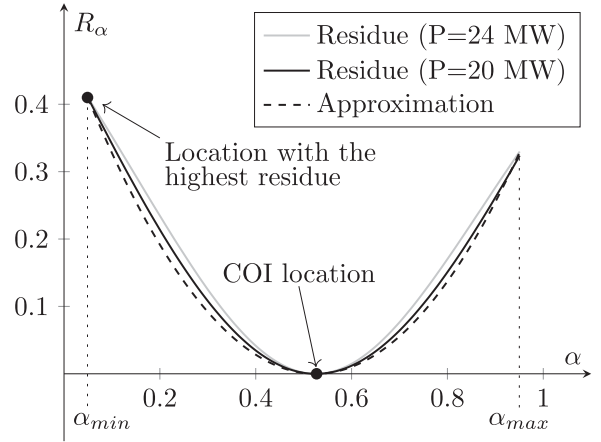


Fig. 3. Comparison between the exact residue against the proposed approximated expression for the 2-machine test system.

one obtained by (19):

$$\alpha^* = \frac{1}{1 + \frac{E'_2}{E'_1} \frac{H_1}{H_2}} \approx \frac{1}{1 + \frac{V_5}{V_1} \frac{H_1}{H_2}} = 0.525 \quad (44)$$

It is worth emphasizing that, although several governor models have been considered such as the IEEE SG0 and IEEE GOV1, they have no impact on the resulting value for α^* . For validation of the residue, modal analysis is performed to obtain eigenvectors and, afterwards, residue is calculated as $R_\alpha = C_\alpha v_\alpha w_\alpha^T B_\alpha$ for a given location α . The residue calculation is also obtained for an even higher power transfer of 24 MW, with a corresponding equilibrium point for angles of $\delta_1 = 72.6^\circ$ and $\delta_2 = 0^\circ$. The comparison of the residue against the approximated expression given by (43) is presented in Fig. 3. For the sake of clarity, this figure only shows the residue for the case of a 6th order machine model and a SCR exciters (the residue for other cases lies within a margin of 2% to the one shown here). In regards to the loading, with 24 MW of power transfer, grid nonlinearities plays a more important role in the system dynamics; still, the approximated and exact residues have a very good agreement. These outstanding results validate the previous theoretical derivations and supports the idea of correlating the residue at a given location for EIR deployment with respect to the distance to the COI location, or in other terms, the distribution of inertia throughout the system. In the next section, based on these analytical derivations, an index of distribution of inertia is proposed which can be employed to identify ideal locations to deploy EIRs in large scale systems.

III. EVALUATION ON LARGE SCALE SYSTEMS

The characteristics of large scale systems are overwhelmingly many and their dynamics complex due to the existence of multiple electromechanical oscillation modes, numerous power flow paths, irregular distribution of inertia throughout the grid, among several other aspects. By focusing on one particular inter-area mode, the dynamic complexity is reduced; still, dynamic behavior dependency on parameters and operational conditions are hard to estimate; explicit expressions for COI location and residue in terms of α are ruled out. However, using the practical concepts described in the previous section, an index associated with the system distribution of inertia is proposed to estimate the electrical distance of any bus to the COI. As being employed to identify ideal locations for EIRs, this index must be validated and statistically correlated with the residue; this is done in a real system, under several operating conditions.

A. Inertia Distribution Index (IDI)

Consider that after a particular perturbation, at time t_0 , the frequency of the COI at any given time $t > t_0$ is estimated as:

$$f_{COI}(t) = \frac{\sum_{i=1}^{n_g} H_i f_i(t)}{\sum_{i=1}^{n_g} H_i} \quad (45)$$

where n_g is the total number of generators, and H_i and $f_i(t)$ are the inertia and measured frequency at time t of generator i . By comparing the frequency of a particular bus k with (45), the distance between the bus and the COI location can be quantified over a time period T as:

$$d_k = \int_{t_0}^{T+t_0} (f_k(\tau) - f_{COI}(\tau))^2 d\tau \quad (46)$$

By normalizing d_k with respect to the highest distance, the following index is defined:

$$IDI_k = \frac{d_k}{\max_{k \in \{1, \dots, n\}} d_k} \quad (47)$$

where n is the total number of buses. The index IDI_k stands for the inertia distribution index at bus k . High values of IDI_k reflects both: (i) the electrical distance from bus k to the COI is relatively large, and (ii) bus k is located in an area with a low concentration of inertia. As the COI location may not be exactly located at a particular bus, the closest bus to the COI is determined as:

$$k^* = \arg \min_{k \in \{1, \dots, n\}} IDI_k \quad (48)$$

To facilitate the application and interpretation in the test system, consider that the index IDI_k is sorted in ascending order and divided into layers; the width of each layer must be properly defined based on the system characteristics—expert knowledge of the system is required. In this fashion, the first layer will include the closest buses to the COI, and the last layer will include the furthest buses from the COI—ideal places to deploy EIRs.

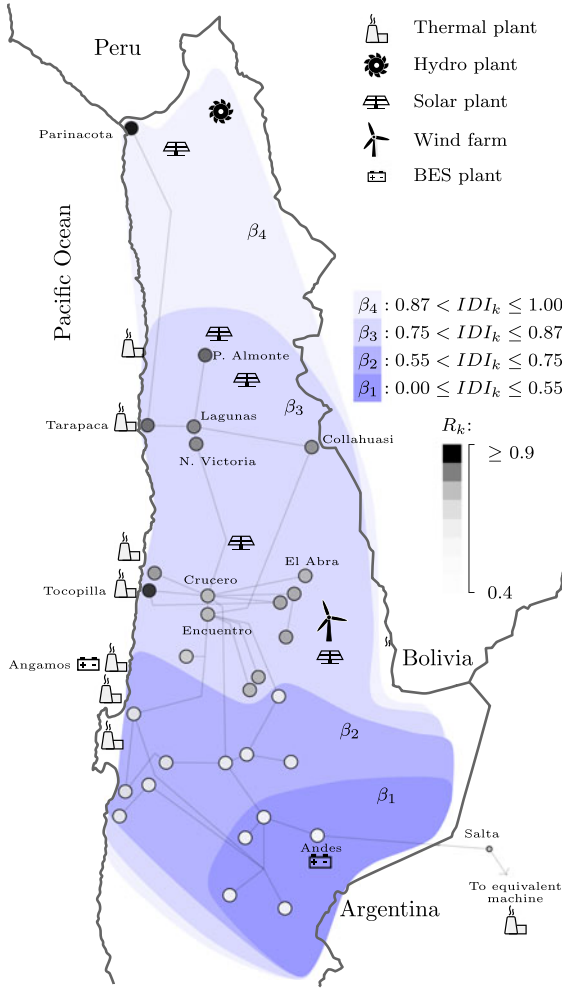
B. Northern Chile Interconnected System (NCIS): A Real Case

This mainly thermal system has 4150 MW of installed capacity, 2400 MW of maximum demand and an approximated

H-constant of 3.86 s based on its installed power. Although the electric consumption remains practically constant—90% attributed to large mining companies—renewable generating systems create power imbalances that will be intensified in the near future due to the large solar energy potential in the area. The system has currently a 90 MW wind farm and several solar power plants totaling 450 MW, none of them equipped with damping or inertia emulation controllers; thus, the impact of renewable generation on the system dynamics is merely through changes in the system operating point. The particular characteristics of the system create favorable conditions for large frequency excursions. To overcome these problems, two Battery Energy Storage (BES) systems of 12 MW and 20 MW were incorporated in 2009 and 2011, respectively [18]. In addition, the NCIS has been interconnected to the Argentinian Interconnected System (AIS) since 2015 to export the generation surplus. The AIS has a total H-constant of inertia of 6.77 s based on an installed power of 31 000 MW, or alternatively, 87.5 s based on the NCIS installed power. As a result, frequency quality and voltage profiles have been improved considerably. However, low damped oscillations between the two systems emerge as a critical problem if no proper action is taken. For a high demand scenario with a transfer of 90 MW from the NCIS to the AIS, the inter-area mode exhibits a critical damping of 0.53%. The system is modeled using a 6th order representation for generators, diverse exciters, governors and power system stabilizers, among other realistic components. The power profile for loads and conventional generation dispatch are defined in three scenarios named high, medium and low demand scenarios. These scenarios and other system data are described in [4], [19], [20]. DIGSILENT Power Factory is used for modeling and simulation with about 1400 state variables and 2700 algebraic variables.

C. Numerical Evaluation

1) *Base Case:* High demand scenario without power production from the renewable generating systems is defined as the base case. To have a first estimate of the index IDI , the system is perturbed by a 100 MW dummy load, step-wisely connected for a time of 0.1 s at Tocopilla bus; later, other locations for this perturbation are also evaluated. Note that dynamic information is needed to obtain IDI ; in a real system, PMU measurements can be used to calculate and keep the IDI values updated. Four layers are employed to identify potentially attractive areas to deploy EIRs, i.e., $\beta_i \forall i \in \{1, 2, 3, 4\}$. Note that, for the sake of simplicity, the calculations are done only at 220 kV buses. The obtained results for R_k , IDI_k and β layers are shown in Fig. 4. The COI is located on the AIS side due to its high inertia in comparison to the NCIS. Note that the distance to the COI gradually increases from south to north, from layer β_1 to layer β_4 which is the furthest area from the interconnection. Observe that the layers are able to group properly the prospective locations for EIR deployment based on their residue, e.g., while layers β_1 and β_2 group locations with a very small residue (light gray circles in Fig. 4), layers β_3 and β_4 capture locations with the highest residue (black circles). Thus, rather than performing heavy calculations related to modal analysis and time-domain simulations to identify prospective locations for EIRs, the proposed index IDI_k can be used, which is simple and effective, and it captures the inertia distribution throughout the system.

Fig. 4. IDI_k , β layers and residue map.

Although further testings and development are necessary to obtain an index not just for radial but also for meshed systems, these results are the first steps towards that final goal. For final verification, full order time-domain simulations and several other operating conditions as well as perturbation locations used to estimate IDI are evaluated.

2) *Time-Domain Simulations*: Consider the base case and assume a short circuit in Crucero busbar which is cleared after 64 ms. Two different locations for an EIR are considered: Andes busbar in layer β_1 , and Tarapaca busbar in layer β_3 . A Flywheel Energy Storage System (FESS) with a damping controller is employed as EIR—for modeling details see reference [4]. Fig. 5 shows the generator speeds after the disturbance. When the FESS is installed in layer β_3 the oscillation is successfully damped in less than 10 s. As a matter of fact the damping ratio is increased to 13.1%. On the other hand, when the FESS is installed in layer β_1 , the oscillation still persists after 10 s exhibiting only a damping ratio of 6.0%.

3) *Impact of Renewable Generation*: As mentioned before, renewable generating systems in the NCIS operate in open-loop, exhibiting no response against frequency disturbances; their only impact on the system dynamics is through changes in their power production, which in turn changes the system operating point. As shown in a previous article [4], using the three load scenarios and assuming that the generation from the

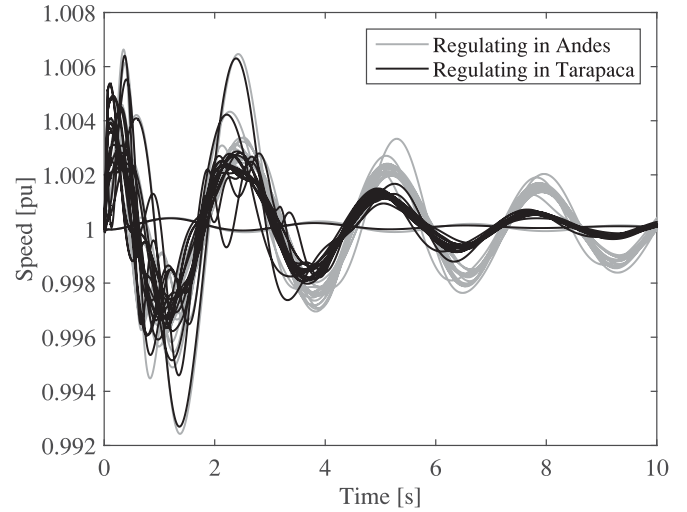


Fig. 5. Generators speed after a 64 [ms] short circuit in Crucero busbar.

solar plants and wind farm goes from 0 to 100% in steps of 20% of their nominal power, a total of 108 operating scenarios are created which should realistically represent possible operational points in the NCIS. The values for $R_{k,s}$ and $IDI_{k,s}$ are calculated for all prospective locations k , and for all operating scenarios s . To evaluate the influence of the location of the 100 MW dummy load on the $IDI_{k,s}$ calculation, all 220 kV busbars on the NCIS/AIS are considered to perturb the system. For visualization purposes, scatter plots of the pairs $(R_{k,s}, IDI_{k,s})$ for only two locations of the dummy load, one at Tocopilla busbar and another at Salta busbar (on the AIS side), are shown in Fig. 6. The Pearson correlation coefficient between R and IDI has been used, which is defined as:

$$\rho_{R,IDI} = \frac{\text{cov}(R, IDI)}{\sigma_R \sigma_{IDI}} \quad (49)$$

Note that $|\rho_{R,IDI}| \leq 1$. In practical terms, when $\rho_{R,IDI}$ approaches +1, R and IDI exhibit a positive linear correlation, which means that IDI increases in the same proportion as R . If $\rho_{R,IDI}$ tends to -1 , then R and IDI have a negative linear correlation, which means that the increment in IDI is in the same proportion as the decrement in R . If $\rho_{R,IDI}$ tends to zero, no correlation whatsoever exists between IDI and R . The Pearson correlation coefficient is approximately equal to 0.919 and 0.927 when the perturbation occurs at Tocopilla and Salta busbars, respectively. The correlation is between 0.9 to 0.95 for all other perturbation locations. Having this coefficient very close to one reflects the highly linear correlation between R and IDI , moreover, this correlation is not sensitive to perturbation location. This validates the idea of using IDI to identify attractive locations for EIR deployment, which is effective, simple, and does not involve heavy mathematical calculations as those in modal analysis.

As a final remark, the decision of whether enabling or not the wind farm or solar plants with damping or inertia emulation controllers should be based on: (a) the trade-off between wind turbines wear and tear and a much higher damping of the inter-area mode, and (b) the availability of energy storage in the solar plants and their capabilities to withstand the transient response of adding either damping or inertia. These power sources are

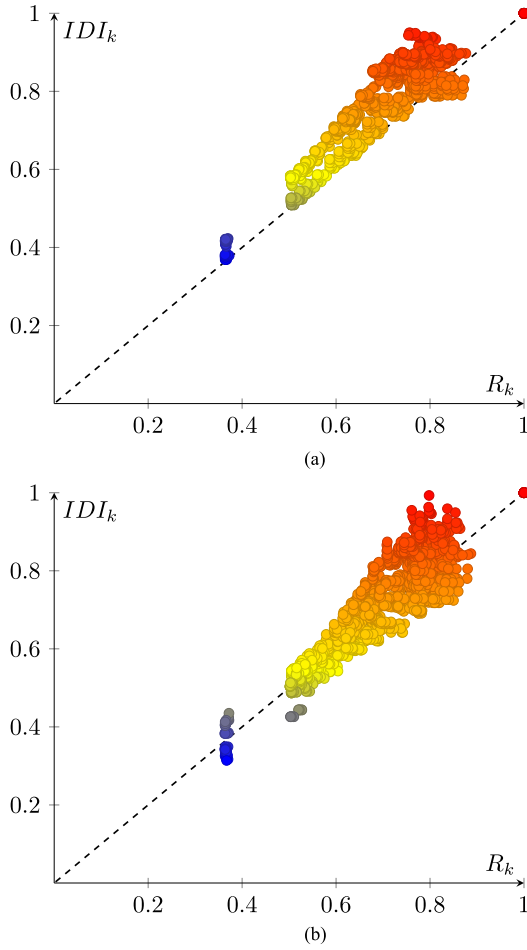


Fig. 6. Scatter plot of the pair $(R_{k,s}, IDI_{k,s})$ for all prospective locations k and evaluated operating scenarios s : (a) $IDI_{k,s}$ calculated after a perturbation at Tocopilla busbar, (b) $IDI_{k,s}$ calculated after a perturbation at Salta busbar.

located in layers β_3 and β_4 , thus, enabling them with these controllers seems to be an attractive choice. Also, the two BES systems can potentially be enabled with these control actions; however, this is not recommended because the damping ratio improvement would be marginal as these BES are located in layers β_1 and β_2 . A simple index as the IDI can help to ease the integration of EIRs and it can be an effective tool for system operators and planners to strength power system dynamics for increased renewable energy.

IV. CONCLUSION

The relationships between oscillations, distribution of inertia and placement of EIRs have been studied. In the case of a 2-machine system, an analytical expression for the physical location of the COI has been established—this expression depends on both H-inertia constant and voltage set-point of the machines. Besides, considering an EIR that may be located anywhere between the two machines, an approximated analytical expression has been derived for the residue of the system transfer function related to the inter-area mode—the larger the residue, the higher the EIR impact displacing the inter-area mode. This expression for the residue is convex in terms of the location and has a minimum value exactly at the COI location. Thus, the basic idea

of locating EIRs further away from the COI, or equivalently, locating EIRs in areas of low inertia, is founded. Based on these analytical derivations, for large scale systems, the index IDI_k for each bus k is proposed which measures the difference square of the bus k frequency with respect to the COI frequency. This index is tested in the NCIS which exhibits very critical inter-area oscillations while interconnected to the AIS. The results obtained by sensitivity analysis, time-domain simulations, and modal analysis validate the application of IDI_k for identifying the best places to deploy EIRs. A total of 108 operating scenarios are evaluated, and a highly linear correlation between IDI and the residue is found. Although more testings and the definition of similar indices for meshed systems with several inter-area modes are needed, the proposed index seems promising to better plan the reinforcement of power systems through EIRs.

APPENDIX

An explicit expression for the electromechanical eigenvalue can be obtained by representing the system through a second-order model. The SM is used as a reference for reduction purposes. Define the following new variables:

$$\begin{aligned}\Delta\omega_1 &= \omega_1 - \omega_2 & \Delta\delta_1 &= \delta_1 - \delta_2 \\ \Delta\omega &= \omega - \omega_2 & \Delta\theta &= \theta - \delta_2\end{aligned}\quad (50)$$

By subtracting the differential equations of the SG and the SM, the following reduced-order model is obtained:

$$\begin{aligned}\Delta\dot{\omega}_1 &= \frac{P_{m1}}{2H_1} + \frac{P_{m2}}{2H_2} - \frac{1}{2H_1} \frac{E'_1 V}{\alpha X} (\Delta\delta_1 - \Delta\theta) \\ &\quad - \frac{1}{2H_2} \frac{V E'_2}{(1-\alpha)X} \Delta\theta\end{aligned}\quad (51)$$

$$\Delta\dot{\delta}_1 = \omega_s \Delta\omega_1 \quad (52)$$

$$0 = -P + \frac{V E'_1}{\alpha X} (\Delta\theta - \Delta\delta_1) \quad (53)$$

$$0 = \frac{V^2}{\alpha(1-\alpha)X} - \frac{V E'_2}{(1-\alpha)X} - \frac{V E'_1}{\alpha X} \quad (54)$$

The linearized zero-input model around the equilibrium point given by $\Delta\omega_{10}$, $\Delta\delta_{10}$, V_0 and $\Delta\theta_0$, becomes:

$$\begin{bmatrix} \dot{z} \\ 0 \end{bmatrix} = \begin{bmatrix} J_1 & J_2 \\ J_3 & J_4 \end{bmatrix} \begin{bmatrix} z \\ y \end{bmatrix} \quad (55)$$

where $z = [\Delta\omega_1 - \Delta\omega_{10}, \quad \Delta\delta_1 - \Delta\delta_{10}]^T$, $y = [V - V_0, \quad \Delta\theta - \Delta\theta_0]$ and

$$J_1 = \begin{bmatrix} 0 & \frac{-V_0 E'_1}{2X\alpha H_1} \\ \omega_s & 0 \end{bmatrix} \quad (56)$$

$$J_2 = \begin{bmatrix} \frac{E'_1(\Delta\theta_0 - \Delta\delta_{10})}{\alpha H_1} - \frac{E'_2\Delta\theta_0}{(1-\alpha)H_2} & \frac{V_0 E'_1}{\alpha H_1} - \frac{V_0 E'_2}{(1-\alpha)H_2} \\ 2X & 2X \\ 0 & 0 \end{bmatrix} \quad (57)$$

$$J_3 = \begin{bmatrix} 0 & \frac{-V_0 E'_1}{\alpha X} \\ 0 & 0 \end{bmatrix} \quad (58)$$

$$J_4 = \begin{bmatrix} \left(\frac{E'_2}{(1-\alpha)} + \frac{E'_1}{\alpha} \right) \Delta\theta_0 - \frac{E'_1 \Delta\delta_{10}}{\alpha} & \frac{V_0 E'_2}{(1-\alpha)} + \frac{V_0 E'_1}{\alpha} \\ X & X \\ \frac{2V_0}{\alpha(1-\alpha)} - \frac{E'_1}{\alpha} - \frac{E'_2}{(1-\alpha)} & 0 \\ X & 0 \end{bmatrix} \quad (59)$$

By Kron's reduction $\dot{z} = Jz$, where:

$$J = J_1 - J_2 J_4^{-1} J_3 \quad (60)$$

$$= \begin{bmatrix} 0 & \frac{V_0 E'_1}{2X\alpha} \left\{ \frac{(1-\alpha)\frac{E'_1}{H_1} - \alpha\frac{E'_2}{H_2}}{(1-\alpha)E'_1 + \alpha E'_2} - \frac{1}{H_1} \right\} \\ \omega_s & 0 \end{bmatrix} \quad (61)$$

In this reduced-order model, J has a pair of purely imaginary conjugate eigenvalues related to the electromechanical oscillations. By solving $|J - \lambda_\alpha I| = 0$, the positive eigenvalue becomes:

$$\lambda_\alpha = \sqrt{\frac{\omega_s V_0 E'_1}{2\alpha X} \left\{ \frac{(1-\alpha)\frac{E'_1}{H_1} - \alpha\frac{E'_2}{H_2}}{(1-\alpha)E'_1 + \alpha E'_2} - \frac{1}{H_1} \right\}} \quad (62)$$

$$= j \sqrt{\frac{\omega_s V_0}{2X} \left(\frac{H_1 + H_2}{H_1 H_2} \right) \frac{E'_1 E'_2}{(1-\alpha)E'_1 + \alpha E'_2}} \quad (63)$$

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